

SkiWalkerVR: Towards Naturalistic Virtual Walking Using Hand Movements and Foot Vibrations

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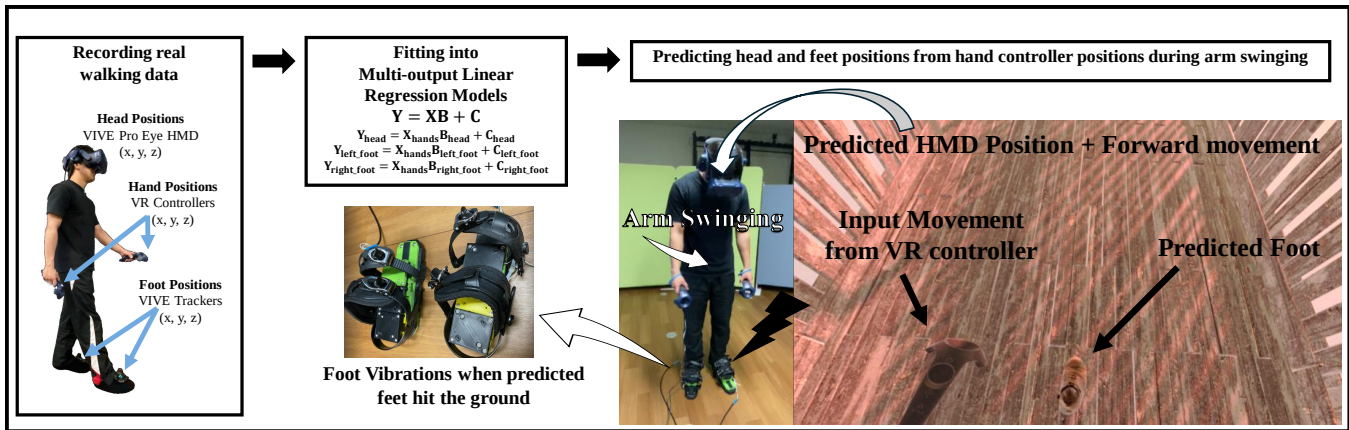


Figure 1: Predicting head and foot positions from hand controllers with multi-output linear regression.

Abstract

Developing immersive Virtual Reality (VR) systems that simulate natural walking experiences in constrained physical spaces remains a challenge. This paper introduces SkiWalkerVR, a novel VR walking system that predicts and replicates head and foot movements based on user-specific hand gestures, creating a naturalistic walking illusion while the user remains otherwise stationary. By integrating synchronized foot vibrations, SkiWalkerVR enhances sensory feedback, increasing the realism of virtual locomotion. We discuss the system's design, implementation, and possible applications, which may include VR walking with improved immersion and reduced motion sickness compared to traditional VR walking methods.

CCS Concepts

• Human-centered computing → Virtual reality.

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Keywords

Virtual walking, locomotion, arm swinging

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1 Introduction

Walking is a fundamental requirement in many Virtual Reality (VR) applications, providing users with a sense of presence and immersion. However, two main challenges must be overcome to enable natural walking in VR, including: (1) support for unconstrained walking in virtual environments larger than the physical tracking space, and (2) providing appropriate multi-sensory stimuli in response to user interactions with the virtual environment [Nilsson et al. 2018].

To address the first challenge, one of potentially multiple solutions such as repositioning systems, redirected walking, and locomotion-based proxy gestures have been proposed. Research has demonstrated that natural experiences can be achieved using upper-body proxy gestures alone, without explicit leg movement.

The method of arm swinging has demonstrated potential for simulating the sensation of walking [McCullough et al. 2015][Nilsson et al. 2013]. Moreover, the results of a study performed by Nilsson et al. (2013) suggest that users experience this gesture to be just as natural as Walking in place (WIP) locomotion and less fatiguing [Nilsson et al. 2013]. A possible reason this gesture was perceived as relatively natural is that it involves a rhythmic swinging of the arms similar to that occurring during real walking [Zehr and Haridas 2003].

In this paper, we expand on the idea of arm swinging by integrating a multi-output linear regression model to predict user-specific head and foot movements based on hand movements to generate a more naturalistic virtual walking experience. This method especially provides active control of virtual walking movements coupled with naturalistic subtle head movements (also active) to the HMD view. Therefore based on previous studies showing that active movements of a virtual avatar reduce motion sickness [Chen et al. 2012], we expect our method to provide a more comfortable walking experience compared to other commonly used methods for locomotion in VR such as moving with button-press, teleportation, and even other previous algorithms that use arm swinging. Additionally, we incorporated vibrotactile shoes that provide foot vibrations when the predicted feet touch the ground to further enhance the walking sensation and immersion based on previous work [Matsuda et al. 2021][Kitazaki et al. 2019]. Therefore, overall our approach aims to offer a more naturalistic walking experience in VR.

2 Design and Implementation

Figure 1 demonstrates the implementation steps. First we record real walking data of the participant. A VIVE Pro Eye head mounted display (HMD), two VIVE hand controllers, and two VIVE trackers were used to track the head, hand and feet positions, respectively in a virtual environment developed using Unity 2022.3.8f1. The participant was asked to walk along a straight line naturally in several speeds to generate a diverse range of training data. The data were recorded at a sample rate of 50 Hz. The recorded data were then fitted into three multi-output linear regression models to predict the head, left foot and right foot positions with respect to the hand positions using Scikit-learn machine learning library for Python. The coefficients and intercepts of the generated linear regression equations were then imported to Unity and the regression equations were executed to predict the x, y, and z coordinates of the head and feet in real-time when the hand coordinates are given.

HMD camera positions in the virtual environment are overwritten with the predicted head positions during arm swinging. This provides the participants with camera (head) movements similar to real walking, even though they are only moving their arms back and forth. However, to minimize cybersickness, we added the real head movements of the participant as well to the HMD position. Rotations of the HMD camera were left as participant's real-time head rotations allowing them to look around freely in the virtual environment. We allow the user to switch between walking mode and stationary mode with a button press on the VR controller so that they can use their hands for other interactions when not walking. In the walking mode, the walking speed is calculated by the

moving average of the speeds of the two hands in a one second time interval multiplied by a multiplier coefficient that the user can adjust for a preferable speed range. This allows the participants to control the walking speed in a similar manner to real walking.

During arm swaying, participants were asked to wear a pair of specially designed vibrotactile shoes as shown in Figure 1. Vibrotactile feedback were presented at left and right forefeet and heels using four vibro-transducers (Acouve Lab Vp408) when the corresponding predicted foot hits the ground in the virtual environment. The transducers were affixed to a 3D-printed ABS plate and supported by four springs, with four units used for each forefoot and heel. An additional unit was used to support the middle foot of the participant (without carrying a transducer). The transducer units were mounted on the modified exterior of ski boots. The transducers were powered by a multichannel power amplifier (Behringer EPQ304, 40W × 4 / 8Ω) and an audio interface, with ASIO utilized for the driver interface. The amplitude of vibration was maintained at a fixed level throughout the demonstration.

3 Discussion

While arm swinging is an existing technique, it is not widely implemented in VR applications currently available. We demonstrate the potential of enhancing virtual walking with arm swinging using machine learning based techniques to predict user-specific walking movements based on hand gestures. Our method is expected to reduce cybersickness due to naturalistic-head movement predictions while allowing users to engage in a more naturalistic form of virtual locomotion in constrained tracking spaces. While this demonstration is based on multi-output linear regression trained on position data of real walking, more complex deep learning models trained on a wide variety of walking data such as limb orientations, speeds, accelerations and jerks etc. as well can be expected to further enhance the virtual walking experience in the future.

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